

Linked List Complete Notes

Data Structures | Unit 1: Arrays and Linked Lists

A linked list stores elements in nodes; each node holds data and a pointer to another node.

- Singly linked list: each node points to the next node; traversal is one-directional.
- Doubly linked list: each node has next and previous pointers, so traversal works both ways.
- Circular linked list: the last node points back to the first node.
- Insertion at the head is $O(1)$; insertion at the tail is $O(n)$ unless a tail pointer is kept.
- Searching a linked list is $O(n)$ because nodes must be visited one by one.
- Compared with arrays: no shifting on insert/delete, but no random access by index.

CampusNotes demo sample - short placeholder content for the hackathon demo.